GIK INSTITUTE — CAPSTONE PROJECT 2026

EduQuery AI

A Smart Question-Answering & Prediction System for School Academic Data

Plain-Language Guide for University Leadership

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Rs. 0	92%	12	4
External Cost (Open-Source Stack)	AI Prediction Accuracy (Research)	Types of Questions It Can Answer	Student Developers Building It

■ What is this document?

This brief explains the EduQuery AI project in plain everyday language — no computer science background required. It is written specifically for university leadership and faculty from non-technical disciplines.

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1 What Is EduQuery AI?

EduQuery AI is a **computer program** that lets school administrators, teachers, and principals **ask questions about student performance in plain English** — exactly the way you would ask a question to a colleague — and receive clear, accurate answers instantly.

■ Think of it like this:

Imagine you have a very smart assistant sitting next to you who has read every mark sheet, attendance record, and exam result from every school in your network over the past several years. You can simply ask: *"Which schools had the lowest pass rate in Physics this year?"* or *"Show me students who missed more than 10 days and are at risk of failing."* The assistant gives you the answer in seconds — no waiting for an IT department, no custom report request, no Excel file.

Additionally, EduQuery AI can **predict future exam results**. Based on a student's history, attendance patterns, and class trends, the system can forecast whether a student is likely to pass or struggle in their next examination — giving teachers time to intervene before it is too late.

At a Glance

Full Name	EduQuery AI: A Prompt-Based Intelligent Analytics and Prediction Engine
What it does	Answers questions about school data in plain English; predicts future exam results
Who uses it	School principals, teachers, administrators — no technical knowledge required
Data source	Existing school database (already in use at participating institutions)
Cost to institute	Near-zero — built entirely on free, open-source software tools
Project type	Student capstone project — Advanced AI Bootcamp, GIK Institute 2026

2 The Problem Being Solved

Schools and school networks collect enormous amounts of data every single day — marks, attendance, teacher performance, exam schedules, and more. However, most of this data **sits unused** because:

✔ Getting a specific answer requires an IT person to write code — this takes days or weeks.

✔ Most software only offers **fixed reports**; you cannot ask a new question that was not pre-designed.

✔ Decision-makers rarely have the technical skills to query databases themselves.

✔ At-risk students are identified **too late** — often after exams are already over.

■ A familiar parallel:

Imagine a bank that stores your complete financial history but can only show you three pre-printed statements — 'Monthly Balance', 'Last 5 Transactions', and 'Annual Summary'. If you want anything else — like 'which months did I overspend?' — you have to call a programmer and wait a week. EduQuery AI turns that bank into a **knowledgeable advisor** who answers any question you have on the spot.

The AI in Education market is **one of the fastest-growing technology sectors globally**. In Pakistan alone, the educational technology market was valued at approximately **$408 million in 2025** and is expected to reach **$2.8 billion by 2034**. Yet no local system offers this kind of intelligent, language-based querying for school academic data. EduQuery AI would be a first of its kind in Pakistan.

3 How It Works — The Simple Version

You do **not** need to understand the technical details to appreciate how EduQuery AI works. Here is the entire process in five simple steps:

1

Step 1 — You Ask a Question
You open a simple website and type your question in plain English, just like sending a WhatsApp message.

2

Step 2 — The AI Reads Your Question
The AI reads your words and figures out what you are really asking — which school, which subject, which time period, which students.

3

Step 3 — The AI Searches the Database
Behind the scenes, the AI translates your question into a search instruction that the school database understands. You never see this — it happens in under 3 seconds.

4

Step 4 — The AI Finds the Answer
The system retrieves the exact data from the database — marks, attendance records, teacher performance — whatever your question needed.

5

Step 5 — You Get a Clear Answer
The answer is displayed as a clear table, chart, or written summary. No jargon. No code. Just the answer.

■ Safety First — The System Cannot Change Anything

EduQuery AI can only READ data — it cannot add, delete, or modify any record in the database. It is like having a read-only window into your data warehouse. Additionally, every single question and answer is logged in an audit trail, so leadership always knows who asked what.

4 What Questions Can It Answer?

EduQuery AI can handle **12 categories of questions**. Here are real examples written in the kind of plain language anyone can use:

Type of Question	Example (just type this in plain English)
Student Lookup	"Show me the profile and contact details of student Roll No. 12345."
Individual Results	"What were Ali Khan's marks in Mathematics for the Midterm 2025?"
Class Performance	"What is the average pass percentage of Class 9 across all schools?"
School Comparison	"Which school had the highest pass rate in English Annual exams?"
Teacher Performance	"List teachers whose students scored below the board average in Physics."
Board Benchmarks	"Which subjects is School A consistently below the national board average?"
Attendance Tracking	"Which students missed more than 15 days before the final exam?"
Exam Operations	"What is the exam schedule for Annual 2026 at GIKI Campus?"
Data Quality Checks	"Are there any students with missing marks in the Midterm results?"
Trends Over Time	"Has performance in Class 10 Science improved or declined over 3 years?"
Future Predictions	"Predict the likely final exam result for Roll No. 5522 in Physics."
Explanations / Why?	"Why is Class 10 at School B performing poorly this year?"

■ Key Point

You do not need to know the technical name of any database table or column. You simply type your question the way you would ask a colleague. The AI handles all the technical translation behind the scenes.

5 Predicting the Future — The AI Brain

One of the most powerful features of EduQuery AI is its ability to **predict future exam results** before those exams happen. This gives teachers and administrators the chance to help struggling students while there is still time.

■■ Think of it like a weather forecast:

A weather forecast does not guarantee rain tomorrow — but it gives you a reliable, evidence-based estimate based on patterns: temperature, humidity, wind speed, and historical data. Similarly, EduQuery AI analyzes patterns in a student's history to estimate their future performance — with a confidence level included, so you always know how certain the prediction is.

What the Prediction Engine Looks At

✔ **Past exam marks** — How did this student perform in previous midterms and annuals?

✔ **Attendance record** — How many days did the student miss before this exam?

✔ **Class context** — How is the overall class performing compared to previous years?

✔ **Teacher performance** — What is this teacher's historical pass rate for this subject?

✔ **Board benchmark gap** — Is this school consistently below or above the national average?

✔ **Trend direction** — Is this student improving, declining, or staying the same over recent exams?

What the Prediction Tells You

Predicted Percentage	An estimate of the student's likely score (e.g., '62–68% likely')
Grade Band	A, B, C, D, or F — the probable grade classification
Pass / Fail Probability	The likelihood of passing or failing (e.g., '78% chance of passing')
At-Risk Flag	A ranked list of students needing urgent attention
School-Level Forecast	Projected class average and school-wide pass rate for planning
Explanation	The AI always explains WHY it made this prediction in plain language

■■ Important: Predictions are Guidance, Not Judgments

Every prediction is labelled as decision support for educators — it is a tool to help teachers identify students who need extra attention. No student's grade, promotion, or future will ever be decided by the AI alone. The final judgment always rests with the teacher and administration.

How Accurate Is It?

A published academic research study conducted in March 2026 demonstrated that the same type of AI used in EduQuery AI achieved **92.32% accuracy** in predicting final letter grades from midterm scores. This benchmark comes from peer-reviewed research published in an international educational data mining journal.

6 Is the Data Safe?

Data security and student privacy are built into EduQuery AI from the ground up — not added as an afterthought. Three independent layers protect the data:

Layer 1 — Login and Roles	Every user must log in with a username and password. A teacher can only see their own school's data. A principal sees their campus. An admin sees everything they are authorized to see. This is enforced by the system — it is not possible to accidentally see another campus's student records.
Layer 2 — Database-Level Lock	Even if a software bug existed (which is guarded against), the database itself has a built-in security rule that prevents any user from seeing data they are not entitled to. This is like a bank vault that only opens for the right keycard — the lock is inside the bank, not just on the front door.
Layer 3 — Sensitive Data Excluded	Passwords, one-time codes, and other sensitive personal information are completely excluded from the AI's view. The AI can never see or expose this data, even if directly asked.
Complete Audit Trail	Every single query, every answer, and every login event is recorded in an audit log. University leadership can review exactly who asked what, and when, at any time.
Read-Only Access	The AI can only READ the database. It cannot add, change, or delete any data. Think of it as a digital window — you can look through it, but you cannot reach through it.

7 Who Built It & How Long Will It Take?

The Student Development Team

EduQuery AI is built by four final-year students at GIK Institute, each specializing in a different aspect of the system:

Name	Role in Plain Language	What They Actually Do
Muhammad Junaid	Language & Database Bridge	Makes the AI understand your English question and translate it into a database search.
M. Zain ul Abideen	Prediction Engine Builder	Trains the AI to predict future exam results by learning from historical patterns.
Abdur Rafay	Backend & Security Architect	Builds the invisible engine room that connects all parts securely and efficiently.
Abeera Anwer	Interface & Experience Designer	Designs the website where users type questions and see answers — making it easy for everyone.

12-Week Development Timeline

Phase	Timeline	What happens in plain language
Phase 1	Weeks 1–2	The team studies the existing school database and teaches the AI what every table and column means in plain language. This is like giving the AI a complete glossary.
Phase 2	Weeks 3–5	The core question-answering system is built and tested. Users can ask basic questions about students, marks, and school performance.
Phase 3	Weeks 6–7	Advanced questions are added — teacher performance, attendance patterns, board comparisons, and exam schedules all become available.
Phase 4	Weeks 8–10	The prediction engine is built and connected. The AI can now forecast future exam results and explain its reasoning.
Phase 5	Weeks 11–12	Security is hardened, the system is tested against 200+ real questions, documentation is finalized, and the live demonstration is prepared.

8 What Will Be Delivered at the End?

At the conclusion of the project, the following concrete, usable outcomes will be produced:

Deliverable	What It Means
Working Web Application	A fully functional website where authorised users can type questions and receive instant answers.
Data Dictionary	A complete, human-readable glossary explaining every data field in the school database — useful for the institution beyond this project.
200+ Tested Questions	A library of verified question-answer pairs demonstrating the system's accuracy across all 12 categories.
Trained Prediction Models	AI models trained on historical exam data, ready to predict future outcomes with documented accuracy.
Explanation Reports	Sample AI explanations showing how predictions are made — ensuring teachers understand the reasoning.
Security Audit Report	Documented proof that data privacy rules are enforced, access controls work correctly, and no sensitive data is exposed.
Full Technical Documentation	Comprehensive documentation of the entire system for future maintenance or extension by GIK Institute's IT team.
Live Demonstration	A live walkthrough showing the system answering real questions and generating predictions in front of the evaluation panel.

9 Why Should GIK Institute Care?

This project delivers concrete, practical value to GIK Institute and to Pakistani education at large — both immediately and as a long-term asset.

Faster Decision-Making
Questions that previously took days to answer through IT requests are answered in seconds. Leadership can make data-driven decisions in real time.

Earlier Student Intervention
At-risk students are identified before exams happen, not after. This gives teachers weeks of extra time to provide support.

Fair, Evidence-Based Review
Teacher and school performance is evaluated against board benchmarks using objective data — not subjective impression.

Zero Software Licensing Cost
The entire system is built on free, open-source software. The institution pays nothing for licenses or subscriptions.

Pakistan's First NL2SQL Education Platform
No similar system exists for school academic data in Pakistan. EduQuery AI positions GIK Institute as a national leader in responsible AI adoption.

Reusable Foundation
The system is modular — any school, district, or institution with a similar database can adopt this platform with minimal adaptation.

Student Excellence Showcase
This project demonstrates the quality of GIK Institute's engineering graduates, combining cutting-edge AI research with practical, real-world institutional value.

10 Frequently Asked Questions

Q: Do we need to install any new software or buy new servers?

A: No. EduQuery AI is designed to work with your existing school database and can run on your current infrastructure. No new server purchases are required during the capstone project. All software components used are free and open-source.

Q: Will this replace teachers or IT staff?

A: Absolutely not. EduQuery AI is a tool that helps teachers and administrators — it does not make decisions, it provides information. Teachers remain fully in control of all educational judgments. IT staff are still needed to maintain the database and infrastructure.

Q: Is it safe to connect AI to our student database?

A: Yes. The AI has read-only access — it cannot modify any data. A three-layer security system (login roles, database-level rules, and sensitive data exclusion) ensures that no unauthorized access is possible. Every interaction is logged in an audit trail.

Q: How accurate are the predictions?

A: Research published in 2026 shows this type of AI achieves over 92% accuracy in similar educational prediction tasks. All predictions include a confidence level so users know how certain the estimate is. Predictions are labelled as guidance tools, not final decisions.

Q: Can it work in Urdu?

A: The current version is designed for English questions. Future versions could be adapted for Urdu if the institution requires it.

Q: What happens to the system after the capstone project ends?

A: All code, documentation, and trained models will be delivered to GIK Institute. The institution's IT team can deploy, maintain, and extend the system independently using the provided documentation. The entire system is built on open-source tools with active global communities.

Q: Who owns the data and the system?

A: GIK Institute and the participating schools own all data. The student team develops the software under an academic capstone framework. All intellectual property is documented and transferred to the institution at project completion.

Q: Does this require an internet connection?

A: The system can be configured to work fully on-premise (within GIK Institute's own network) without sending any data to external internet servers. This is the maximum-privacy option and is fully supported.

EduQuery AI

Turning school data into answers.
Turning historical patterns into foresight.
Turning information into action.

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This document is a plain-language companion to the full technical proposal. For the complete technical specification, architecture diagrams, and source references, please refer to the main EduQuery AI Proposal document.